

海上回收着陆论文真实性、准确性与完整性审查

审查编号: SeaLandingPaper-ResearchIntegrity-Audit-v1

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机器审查结论: PASS_WITH_DECLARED_LIMITATIONS

1. 审查目的与判定规则

本文件用于回答三个问题: 正文中的结果是否由本项目代码重新计算, 文献数据是否仅作为明确标注的验证目标, 以及尚未闭合的模型环节是否被如实保留。审查不以‘图能画出来’为通过条件, 而要求计算输入、生成代码、机器可读输出和成稿图之间可以逐项追溯。

本文结果分为四类。present_calculation 是本项目在本机重新执行得到的结果; reference_target 是从文献转录或数字化的验证目标; publishedinput 是模型采用且已注明来源的文献参数; identified/proxy 是由于公开资料缺失而辨识或假设的输入。后两类不得写成原作者公布值, 也不得用于声称专有模型已被完整复现。

2. 主要数值证据

矩形驳船 HAMS 输出包含 19 个频率、7 个浪向和 5 个甲板点。参数化网格与水静力自检全部通过。着陆中心最大垂荡 RAO 为 1.128 m/m, 发生在 0.6 rad/s、90 deg。该数值由本地编译的 HAMS 输出重新计算, 不取自任何参考论文。

随机波筛选包含 63 个海况-浪向组合, 每个组合 100 个确定性随机相位种子, 单次时长 600 s, 步长 0.1 s。全矩阵最大着陆中心 P95 垂向速度为 1.505 m/s, 对应 Hs3_Tp10、90 deg。谱矩归一化和显著波高线性检查来自脚本自动判定。

Chrono-Cummins 代理算例的本机计算峰值接触力为 2.678 MN, 峰值缓冲行程为 46.44 mm。但最大 SMC 贯入为 0.148 m, 占足垫半径的 74.1%。因此这些力学量仅用于证明开放耦合链可执行, 不作为真实型号支腿载荷的验证值。

Yang 对照图中的虚线来自公开试验曲线的像素数字化, 实线来自本项目模型。参数只使用 simultaneous touchdown 工况辨识, 1-2-1 与 2-2 工况没有重新调参。缓冲行程峰值误差较小, 但加速度峰值误差仍为 29.2%-51.9%, 说明公开资料不足以重构原柔性机构。这一差异被保留, 没有通过平滑、缩放或手工修改曲线消除。

3. 数据与文件溯源

文件	证据类别	用途	存在	SHA-256 前缀
新论文参考写作论文 /Analysis+of+Sea-Based+Landing+Dynamics+of+Reusable+Landing+Vehicle+Considering+Mechanism+Flexibility.pdf	reference,source	Yang et al. validation paper	是	e7949d55dea5a5d9

文件	证据类别	用途	存在	SHA-256 前缀
新论文参考写作论文 /Analysis+of+Sea- Based+Landing+Dynam- ics+of+Reusable+Landing +Vehicle+Considering+ Mechanism+Flexibility.m d	reference_source	Yang et al. text extraction	是	3939e6f442f1738d
海上平台火箭回收文献 /065002_1_2.0002061.pd f	reference_source	Nargolkar and Vijayan benchmark paper	是	f9ba2eb6197a8ad7
海上平台火箭回收文献 /065002_1_2.0002061.m d	reference_source	Nargolkar and Vijayan text extraction	是	37bc58992de0399c
RocketRecoveryCases/Pa per_Yang_2026/reference /digitized/yang-2026- fig09-11-digitization.json	reference_target	Digitized published validation traces and raster calibration metadata	是	309fb9da09933f8c
RocketRecoveryCases/Pa per_Yang_2026/yang- 2026-evidence-report.json	present_calculation	Locally generated deck-filter and source-consistency report	是	c00ffc3b55deb655
RocketRecoveryCases/Pa per_Yang_2026/identifie d_landing_model/yang- 2026-identified-landing- report.json	present_calculation	Locally identified and predicted landing responses	是	da1f143fe0d4051e
RocketRecoveryCases/Ba rge_120x50/platform_con fig.json	present_modelinput	Parameter- controlled rectangular-barge case	是	50f90e7fbd6f1f1b
RocketRecoveryCases/Ba rge_120x50/validation/ba rge-self-check.json	present_calculation	Analytic geometry and hydrostatic checks	是	06359d8905263bfc
RocketRecoveryCases/Ba rge_120x50/Output/Rock etRecovery/deck-point- rao.json	present_calculation	Locally compiled HAMS deck-point RAOs	是	c07aad254806fce5
RocketRecoveryCases/Ba rge_120x50/Output/Rock etRecovery/sea-state- response.json	present_calculation	Spectral response integrals from HAMS RAOs	是	fcac3cc88010888d
RocketRecoveryCases/Ba rge_120x50/Output/Rock etRecovery/wave- sensitivity.json	present_calculation	Random-phase HAMS/JONSWAP statistics	是	2284f1c194ce227c
RocketRecoveryCases/Ch rono_LeggedRecovery/ch rono-stage3-two-way- report-data.json	present_calculation_with_proxy inputs	Locally executed Chrono-Cummins mechanism-proxy response	是	4b983d7c6812d0ba

完整哈希值保存在 research-integrity-audit.json。每次重算后重新生成成本审查文件，可以识别输入、输出或图件是否发生变化。

4. 图件审查

主文图统一由 analysis/rocket_recovery/yang_2026_figures.py 生成，使用 SciencePlots 的 science 与 no-latex 样式、Times New Roman 字体，同时输出 600 dpi PNG 和可编辑 SVG。文献曲线采用灰黑虚线或点列，本文计算采用彩色实线。图件不经过 PowerPoint、Illustrator 或图像软件手工移动数据。

图件	数据性质	PNG 像素	SVG 字体含 Times New Roman	SHA-256 前缀
fig01-coupled-framework	Method diagram generated from the implemented data flow; no numerical result is imported.	4299 x 2019	是	8d537aea6dcd17bb
fig02-barge-geometry-mesh	Present parameterized geometry and registered deck points.	2756 x 2259	是	c714c50dad5ec0e
fig03-yang-deck-filter-reproduction	Present deterministic response; dashed spectral shape is reconstructed from the published transfer function.	4299 x 2769	是	50374283dedb022f
fig04-yang-identified-landing-comparison	Dashed curves are published tests digitized for validation; solid curves are present calculations.	4235 x 3159	是	d37ae985bbbd4641
fig05-barge-hams-rao-sea-response	Entirely present HAMS and spectral calculations.	4299 x 2799	是	9403cc0bef572cd6
fig06-chrono-cummins-stage3-proxy	Entirely present calculation using explicitly identified published and proxy inputs.	4299 x 2799	是	702a78abc18e1725
fig07-barge-wave-sensitivity	Entirely present 63-case, 100-seed linear wave-screening calculation.	4299 x 2799	是	b9e45e726f66e809

第一轮图审发现并修正了两类问题：甲板滤波图曾把绝对 Welch 谱与未归一化传递函数平方直接叠加，现改为归一化谱形比较；耦合框架图曾有一条甲板运动箭头方向错误，已按 Cummins→甲板运动→Chrono、Chrono→平台接触力矩的实际接口改正。第二轮检查覆盖图例可见性、单位、峰值、坐标范围和文献/本文线型区分。

5. 尚未解决的真实性问题

严重度	项目	证据	对论文结论的约束
high	Chrono contact compliance	Maximum SMC contact penetration is 0.148 m, or 74.1% of the 0.2 m footpad radius.	Stage 3A contact-force peaks are retained only as mechanism-proxy outputs and are not treated as validated full-scale leg loads.
high	Landing-leg geometry and absorber law	The supplied papers do not publish CAD hinge coordinates, member inertia, or a machine-readable hydropneumatic force-stroke/velocity curve.	The Chrono branch cannot be described as an Adams-equivalent reproduction of Yang or Thies.
medium	Yang model identifiability	The stroke observation ratio reaches its declared upper bound and acceleration peak errors remain 29.2-51.9%.	Only stroke magnitude and broad touchdown-sequence trends are supported by the public-data reduced model.
medium	Strong two-way closure	The accepted Stage 3A calculation uses a one-pass heave/roll/pitch correction; full 6-DOF fixed-point closure remains incomplete.	The paper reports an executable partitioned demonstration, not converged strong coupling.
medium	HAMS legacy benchmark portability	HywindSpar matches the registered benchmark, while Cylinder, DeepCwind, and Moonpool differ under the local GNU Fortran toolchain.	Analytic barge checks and locally reproducible outputs are valid; universal cross-toolchain equivalence is not claimed.

严重度	项目	证据	对论文结论的约束
low	Random-wave interpretation	The 63-case study is a linear platform-only HAMS/JONSWAP screening calculation.	Its P95 values are not landing success probabilities or operational sea-state limits.

6. 可重复执行命令

```
.\tools\paper-env\Scripts\python.exe .\analysis\rocket_recovery\yang_2026.py report
.\tools\paper-env\Scripts\python.exe .\analysis\rocket_recovery\yang_2026_landing_identification.py
report
.\tools\paper-env\Scripts\python.exe .\analysis\rocket_recovery\barge_wave_sensitivity.py report
.\tools\paper-env\Scripts\python.exe .\analysis\rocket_recovery\yang_2026_figures.py
.\tools\paper-env\Scripts\python.exe .\analysis\rocket_recovery\paper_integrity_audit.py report
```

HAMS 与 Chrono 的完整重算分别由项目现有 PowerShell 流水线调用本地编译求解器和 .tools/chrono-env。论文制图环境与 Chrono 环境分离，SciencePlots 的安装不修改系统 Python 或系统 PATH。

7. 审查结论

现有 HAMS 驳船、JONSWAP 筛选、Yang 公开数据辨识模型和 Chrono-Cummins 代理结果均有机器可读输出和生成代码，可以证明主文中的‘本文计算’并非从参考论文复制。文献值只在验证曲线和输入参数中出现，并已明确标注。与此同时，当前模型仍不具备真实支腿 CAD、经试验标定的缓冲器曲线、可接受的全尺度接触贯入和全 6DOF 强耦合闭合，因此论文必须保持为开放中等保真度框架与分层验证研究，不能写成某型火箭海上回收全过程已完成工程验证。